Avash Palikhe

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Profile

Ph.D. student at the Knight Foundation School of Computing & Information Sciences (KFSCIS), Florida International University, advised by Dr. Fahad Saeed. Graduate Research Assistant at the Precision Computational Health and Biomedical Data Science Lab (PCDS Lab, also known as the Saeed Lab). Research interests: machine learning, proteomics, and trustworthy AI.

Education

- **Florida International University** — Miami, USA
  Ph.D. in Computer Science — *August 2025 – Present*
  Doctoral Advisor: *Dr. Fahad Saeed*
  Research Area: *Machine Learning, Proteomics, & Trustworthy AI*

- **Tribhuvan University** — Kathmandu, Nepal
  Bachelor's in Computer Engineering — *July 2018 – April 2023*
  Courses: *Electrical Engineering, Software Engineering, Theory of Computation, Artificial Intelligence, Mathematics, Data Structure & Algorithm, Probability & Statistics, Microprocessors, Thermodynamics, Electromagnetics*

Skills

- **Programming:** Python, Java, C++, Kotlin, Dart
- **ML & Computing:** Deep learning, LLMs, PyTorch, HPC (SLURM), Linux, Git
- **Domain:** LLMs for mass spectrometry-based proteomics

Publications

- Sribala Vidyadhari Chinta, Zichong Wang, Avash Palikhe, Xingyu Zhang, Ayesha Kashif, Monique Antoinette Smith, Jun Liu, and Wenbin Zhang. "AI-driven healthcare: Fairness in AI healthcare: A survey." PLOS Digital Health 4, no. 5 (2025).
- Shamim Yazdani, Akansha Singh, Nripsuta Saxena, Zichong Wang, Avash Palikhe, Deng Pan, Umapada Pal, Jie Yang, and Wenbin Zhang. "Generative AI in depth: A survey of recent advances, model variants, and real-world applications." Journal of Big Data 12, no. 1 (2025): 1–43.
- Zhipeng Yin, Zichong Wang, Avash Palikhe, Zhen Liu, Jun Liu, and Wenbin Zhang. "AMCR: A framework for assessing and mitigating copyright risks in generative models." In Proceedings of the 28th European Conference on Artificial Intelligence (2025).

Preprints

- Avash Palikhe, Zichong Wang, Zhipeng Yin, Rui Guo, Qiang Duan, Jie Yang, and Wenbin Zhang. "Towards transparent AI: A survey on explainable language models." arXiv preprint arXiv:2509.21631 (2025).
- Avash Palikhe, Zichong Wang, Zhipeng Yin, Shu Hu, Xuejiao Zhao, Jun Liu, Yu Cai, and Wenbin Zhang. "Domain Knowledge Empowered Language Models: A Survey." TechRxiv. November 25, 2025.
- Jiale Zhang, Zichong Wang, Avash Palikhe, Zhipeng Yin, and Wenbin Zhang. "Datasets for Fairness in Language Models: An In-Depth Survey." arXiv preprint arXiv:2506.23411 (2025).

Tutorials

- Zichong Wang, Avash Palikhe, Zhipeng Yin, Jiale Zhang and Wenbin Zhang. "Fairness in Large Language Models: A Tutorial." The 34th International Joint Conference on Artificial Intelligence (IJCAI), 2025.
- Zichong Wang, Avash Palikhe, Zhipeng Yin and Wenbin Zhang. "Fairness in Language Models: A Tutorial." The 34th ACM International Conference on Information and Knowledge Management (CIKM), 2025.
- Zhipeng Yin, Zichong Wang, Avash Palikhe, and Wenbin Zhang. "Uncertain boundaries: A tutorial on copyright challenges and cross-disciplinary solutions for generative AI." In Proceedings of the 34th ACM International Conference on Information and Knowledge Management, pp. 6853–6856. 2025.
- Zichong Wang, Avash Palikhe, Zhipeng Yin and Wenbin Zhang. "Fairness in Language Models: A Tutorial." The IEEE International Conference on Data Mining (ICDM), 2025.

Reviewer

- Conferences: NeurIPS 2026
- Journals: International Journal of Human-Computer Interaction.

Experience

- **Florida International University** — Miami, USA
  Graduate Research Assistant — *August 2025 – Present*
  - Developing an LLM-based approach to de novo peptide sequencing from MS/MS spectra using chain-of-thought reasoning over b/y-ion mass ladders.

- **Florida International University** — Miami, USA
  Graduate Teaching Assistant — *January 2026 – Present*
  Courses: *COP 4813 Web Application Programming*

- **Nepal Clearing House Limited** — Kathmandu, Nepal
  Android Engineer — *October 2023 – March 2025*
  Areas of Work: *Payment Systems, Android Development, Flutter, Java, Kotlin, Dart*
  - Built ConnectIPS, an interbank payment app (native Android, 1M+ downloads), and NepalPay Business, a Flutter merchant payments app (50K+ downloads).

- **Ayata Incorporation** — Kathmandu, Nepal
  Intern Android Developer — *January 2023 – March 2023*
  Areas of Work: *Native Android Development, Kotlin, Java*

Honors & Awards

- **Winner, Blackstone Challenge, ShellHacks Hackathon 2026** — Miami, USA
  PortfolioFit*: AI investing app with natural-language chat, generative UI, and visualization* — *September 2026*

Service & Affiliations

- **IEEE FIU Student Branch** — Miami, USA
  Member — *August 2026 – Present*
- **INIT FIU** — Miami, USA
  Member — *August 2026 – Present*
- **Graduate Professional Association at FIU** — Miami, USA
  Member — *August 2026 – Present*